CHAPTER 1 : MEASUREMENT AND ERROR

1.0 INTRODUCTION

Over the last century and a half, there have been many contributions to the art of measuring electrical quantities. During most of this period, the principal effort was aimed at perfecting a deflection-type instrument with a scale and effort and moveable pointer. The angle of deflection of the pointer is a function of, and is therefore analogous to, the value of electrical quantity being measured.

1.1 MEASUREMENT

1.1.1 Definition

Measurement can be define as a process the amount, quantity and degree or capacity physical parameter by comparison it with standard of reference system of units employed.

All measures have a unit attached to them: inch, centimeter, dollar, liter, ohm, volt, etc. In contrast, an indicator is an indirect measure or a predictor (such as a leading Indicator) of performance.

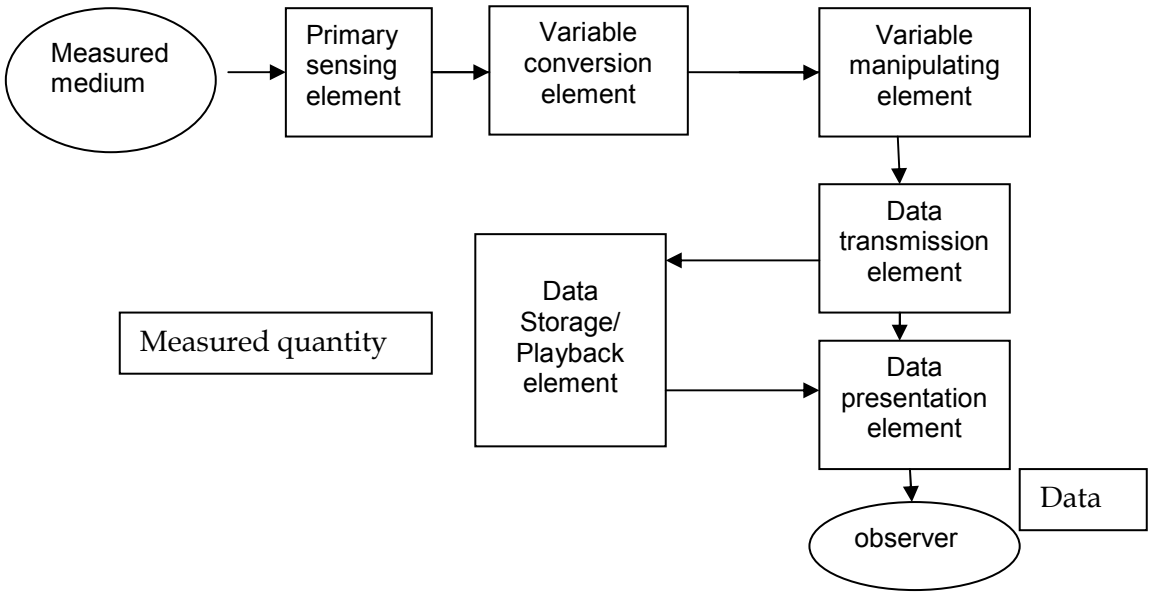


Figure 1.1 : Block Diagram Measurement Process

It involves connecting a measuring instrument into a system under consideration and observing the resulting response on the instrument. The mesurement thus obtained is a quantitative measure of the so-called true value. Since it is very difficult to define the true value adequately, the expected value is used throughout the text.

For example, connecting a measuring instrument into the circuit under consideraton always disturbs (changes) the circuit, causing the measurement to differ from the expected value.

1.1.2 Elements of Measurement System.

i) Primary Sensing Element

This is the element that first receives energy from the measured medium and procedures an output depending un some way on the measured quantity (measured).

The putput is some physical variable e.g displacement or voltage. An instrument always extract some energy from the measured medium. The measured quantity is always disturbed by the act of measurement, which makes a perfect theoritically impossible. Good instruments are designed to minimize this loading effect.

ii) Variable Convertion Element

If may be necessary to convert the output signal of the primary sensing element to another more suitable variable while preserving the information content of the original signal. This element performs this function.

iii) Variable manipulation element

An instrument may require that a signal represented by some physical variable be manipulated by some way.

By manipulation we mean specifically a change in numerical value according to some definite rule but a preservation of the physical nature of the variable. This element performs such a function.

iv) Data-Transmision Element

When functional elements of an instrument are actually physically seperated, it becomes necessary to transmit the data from one to another. This element performs this function.

v) Data presentation element

If the information about the measured quantity is to be communicated to a human being for monitoring, control, or analysis purposes, it must be put into a form recognizable by one of the human senses. This element performs this 'translation' function.

vi) Data Storage/Playback Element

Some applications require a distinct data storage/playback which can easily recreate the stored data upon command.

Note: A given instrument may involve the basic functions in any number, combination, or order. A given physical componenet may serve several of the basic functions.

Quantity	Symbol	Unit	Symbol
Length	l	meter	m
Mass	m	kilogram	kg
Time	t	second	s
Temperature	T	kelvin	^{o}K
Electric current	I	ampere	A
Amount of Substance		mole	mol
Luminous intensity		candela	cd

Table 1: Base Units of Measurement

Quantity	Symbol	Unit	Unit Abbre.
Voltage (emf)	V	volt	V
Charge	Q	coulomb	C
Resistance	R	Ohm	Ω
Capacitance	C	farad	F
Inductance	L	henry	H

Table 2 : Electrical Units

1.1.3 **Terminology**

i. Scale

Definition of device, or procedure used in arranging, measuring, or quantifying events, objects, or phenomenon in any sequence. Scales may be broadly classified as

a) **Category scale**:
Sequence of numbers or words which only serve to identify certain entities or observations and have no quantitative significance; for example, a numbered or named list, or numbers on the uniforms of the members of a team. Also called nominal scale or qualitative scale.

b) Interval scale:
Sequence of numbers in a fixed order representing amount, interval, or numeric values; for example, a distance or temperature scale. Also called quantitative scale.

ii. Range

General definition : Universe of all realistic possibilities.

1.2 TYPE OF ERROR

Errors are generally categorized under the following three major types:

i. Gross error

This class of errors is generally the fault of the person using the instruments such as incorrect reading of instruments, incorrect recording of experimental data or data incorrect use of instruments.

As long as human beings are involved, some gross errors will definitely commit. Although complete eliminating of gross errors is probably impossible, one should try to avoid them. The following actions may be necessary to reduce the effects of gross errors.

a) Great care should be taken in reading and recording the data.
b) Two or more readings should be taken by different experimenters

ii. Systematic error

Systematical errors can be divided into four categories:

a) Instrumental Error

These errors arise due to main reasons:

i. Due to inherit shortcoming in the instruments (may be caused by the construction, calibration or operation of mechanical structure in the instruments).
ii. Due to misuse of the instruments. For example, these may be caused by failure to adjust zero of the instruments.
iii. Due to loading effect of the instruments. These errors can be eliminated or at least reduced by using the following methods:

a) The procedure of measurements must be carefully planned.
b) Correction factors should be applied after detection of these errors.
c) Re-calibration the instrument carefully.
d) Use the instrument intelligently.

b) Observational Errors

i. Due to the types on instrument display, whether it is analog or digital.
ii. Due to parallax (eye should be directly in line with the measurement point).

Note: *These errors can be eliminated completely by using digital display instruments.*

c) Environmental Errors

Due to conditions external to the measuring device such as the area, surrounding the instrument. These conditions may be caused by the changes in pressure, humidity, dust, vibration or external magnetic or electrostatic fields. These errors can be eliminated or reduced by using corrective measure such as:

i. Keep the condition as constant as possible.

ii. Use instrument/equipment which is immune to these effects.

iii. Employ technique which eliminates these disturbances.

d) Simplification Errors

Due to simplification of a formula: For example: A = B + C + D2.
If D is too small, then the formula is simplified to: A = B + C.
There will be a different result between the first and the second equation. In high accuracy requirements, a formula should not be simplified to avoid these types of errors.

iii. Random error

In some experiments, the results shows variation from one to another, even after all systematical and gross errors have been accounted for. The cases of these errors are not recognized, therefore the elimination or reduction of these errors are not possible. When these types of errors are occurred, the best result can be determined by statistical analysis.

1.2.1 Error in Measurement

Error of Measurement can be define difference between the actual value of a quantity and the value obtained by a measurement. Repeating the measurement will improve (reduce) the random error (caused by the accuracy limit of the measuring instrument) but not the systemic error (caused by incorrect calibration of the measuring instrument).

i. Absolute error

Can be defined as the difference between the expected value of the variable and the measured value of the variable.

where

e = absolute error
Y_n = expected value
X_n = measured value

$$e = Y_n - X_n \qquad \text{(Equation 1.1)}$$

c) Relative error (Percent of Error)

To express the error as a percentage,

$$Percent\ error = \frac{absolute\ error}{expected\ error}\ x\ 100\%$$

From equation 1.1 where $Absolute\ error, e = Y_n - X_n$, so

$$Percent\ error = \frac{e}{Y_n}\ x\ 100\%$$

(Equation 1.2)

or

$$Percent\ error = \frac{Y_n - X_n}{Y_n}\ x\ 100\%$$

(Equation 1.3)

Example 1

The expected value of the voltage across a resistor is 50V; however, measurement yields a value of 49V. Calculate

a) The absolute error
b) The percent of error

Solution

a) $e = Y_n - Xn$
$= 50V - 49V$
$= \mathbf{1V}$

b) $Percent\ error = \frac{50-49}{50}\ x\ 100\%$
$= 0.02\ x\ 100\%$
$= \mathbf{2}\%$

1.2.2 Characteristic of Measurement

a) Accuracy

The degree of exactness of a measurement compared to the expected value, or the most probable value, of the variable being measured.

Accuracy expressed as

Percent accuracy, a = 100% - percentage of error
= A X 100% (Equaition 1.4)

b) Precision

A measure of the consistency or repeatability of measurements. Precision can be expressed as:

$$Precision = 1 - \frac{X_n - \overline{X_n}}{\overline{X_n}} \qquad \text{(Equation 1.5)}$$

where : X_n = the value of the nth measurement.
$\overline{X_n}$ = the average of the set of n measurements.

c) Relative

It is frequently more desirable to express measurement in terms of relative accuracy rather than error.

$$Relative\ accuracy, A = 1 - \frac{Y_n - X_n}{Y_n} \qquad \text{(Equation 1.6)}$$

Example 2

Refer the question at Example 1. Calculate

a) The relative accuracy
b) The percent of accuracy

Solution

$$a)\ A = 1 - \frac{50 - 49}{50} = 1 - \frac{1}{50}$$
$$= 1 - 0.2$$
$$= \mathbf{0.98}$$

$$b)\ a = 100\% - percent\ of\ error$$
$$= 100\% - 2\%$$
$$= \mathbf{98}\%$$

Example 3

The following set of ten measuremet was recorded in the liboratary. Calculate the precision of the fourth measurement.

Measurement Number	Measurement Value X_n (volts)
i.	98
ii.	102
iii.	101
iv.	97
v.	100
vi.	103
vii.	98
viii.	106
ix.	107
x.	99

Solution:

The average value for the set of measdurements is equal to the sum of the measurements divided by 10, which 101.1. The precision of the fourth measurement is

$$Precision = 1 - \frac{X_n - \overline{X_n}}{\overline{X_n}}$$

$$= 1 - \frac{97 - 101.1}{101.1}$$

$$= 1 - 0.04$$

$$= \mathbf{0.96}$$

c) Resolution

The smallest change in a measured variable to which a instrument will respond.

d) Significant figure

When making measurement or calculation, we retain only significant figures. Significant figures are the figures, including zeroes and estimated figures, that have been obtained from measuring instruments known to be trustworthy. The position of the decimal point does ot affect the number of significant figures. However, if it actually represents a digit read with an instrument.

Significant Figure Rules

- Non-zero digits are always significant.
- All zeros between other significant digits are significant.
- The number of significant figures is determined starting with the leftmost non-zero digit. The leftmost non-zero digit is sometimes called the *most significant digit* or the *most significant figure*. For example, in the number 0.004205 the '4' is the most significant figure. The lefthand '0's are not significant. The zero between the '2' and the '5' is significant.

- The rightmost digit of a decimal number is the *least significant digit* or *least significant figure*. Another way to look at the least significant figure is to consider it to be the rightmost digit when the number is written in scientific notation. Least significant figures are still significant! In the number 0.004205 (which may be written as 4.205×10^{-3}), the '5' is the least significant figure. In the number 43.120 (which may be written as 4.3210×10^{1}), the '0' is the least significant figure.
- If no decimal point is present, the rightmost non-zero digit is the least significant figure. In the number 5800, the least significant figure is '8'.

Uncertainty in Calculations

Measured quantities are often used in calculations. The precision of the calculation is limited by the precision of the measurements on which it is based.

Addition and Subtraction

When measured quantities are used in addition or subtraction, the uncertainty is determined by the absolute uncertainty in the least precise measurement (not by the number of significant figures). Sometimes this is considered to be the number of digits after the decimal point.

Example 4

Have 3 value of measurement : 32.01 m , 5.325 m, 12 m. State the sum of value should be report.

Solution

When added together, the sum will get 49.335 m, but the sum should be reported as '49' meters.

Multiplication and Division

When experimental quantities are mutiplied or divided, the number of significant figures in the result is the same as that in the quantity with the smallest number of significant figures. If, for example, a density calculation is made in which 25.624 grams is divided by 25 mL, the density should be reported as 1.0 g/mL, not as 1.0000 g/mL or 1.000 g/mL.

Exact Numbers

Sometimes numbers used in a calculation are exact rather than approximate. This is true when using defined quantities, including many conversion factors, and when using pure numbers. Pure or defined numbers do not affect the accuracy of a calculation. You may think of them as having an infinite number of significant figures. Pure numbers are easy to spot because they have no units. Defined values or conversion factors, like measured values, may have units. Practice identifying them!

Example 5:
You want to calculate the average height of three plants and measure the following heights: 30.1 cm, 25.2 cm, 31.3 cm; with an average height of (30.1 + 25.2 + 31.3)/3 = 86.6/3 = 28.87 = 28.9 cm. There are three significant figures in the heights. Even though you are dividing the sum by a single digit, the three significant figures should be retained in the calculation.

Example 6
Three students weigh an item using different scales. These are the values they report:
a. 20.03 g
b. 20.0 g
c. 0.2003 kg
How many significant figures should be assumed in each measurement?

Solution
a. 4.
b. 3. The zero after the decimal point is significant because it indicates that the item was weighed to the nearest 0.1 g.
c. 4. The zeros at the left are not significant. They are only present because the mass was written in kilograms rather than in grams. The values '20.03 g' and '0.02003 kg' represent the same quantities.

In addition to the solution presented above, be advised that the correct answers to have been obtained very quickly by expressing the masses in scientific (exponential) notation:

$20.03\ g = 2.003 \times 10^{1}\ g$ (4 significant figures)
$20.0\ g = 2.00 \times 10^{1}\ g$ (3 significant figures)
$0.2003\ kg = 2.003 \times 10^{-1}\ kg$ (4 significant figures

1.3 STANDARD USED IN THE MEASUREMENT

Universally or widely accepted, agreed upon, or established means of determining what something should be. Major classifications of this term include:

i. Material or substance whose properties are known with a level of accuracy that is sufficient to allow its use as a physical reference in calibrating or measuring the same properties of another material or substance.

ii. Concept, norm, or principle established by agreement, authority, or custom, and used generally as an example or model to compare or measure the quality or performance of a practice or procedure.

iii. Written definition, limit, or rule approved and monitored for compliance by an authoritative agency (or professional or recognized body) as a minimum acceptable benchmark. This is the usual meaning of the plural term (standards). See also specification.

1.3.1 Categorized into four:

a) International standard

Are defined by international agreements. These standards are maintained at the International Bureau of Weight and Measures in Paris. Frances, They are periodically evaluated and checked by absolute measurements in term of the fundamental units of physics. They represent certain units of measurement to the closest possible accuracy attained by the science and technology of measurement and used for comparison with primary standards.

b) Primary standard

The Standard is maintained at institution in various countries around the world, such as the National Bureau of Standard on Washington D.C, SIRIM in Malaysia. The primary standards are not available for use outside the national laboratories. Their principle function is to calibrate and verify the secondary standards.

c) Secondary standard

Used as the basic reference standards used by measurement & calibration laboratories in the industry. Each industrial laboratory is completely responsible for its own secondary standards. Each laboratory sends its secondary standards to the national standards (primary standards) laboratory for calibration. After calibration, the secondary standards are returned to the industrial uses with the certification and checked periodically.

d) Working standard

Working standard is the principle tools of a measurement laboratory and the lowest level of standards. They are used to check and calibrate the instruments used in the laboratory or to make comparison measurement in industrial application. As example, the standard resistor, capacitors and inductor**.**

EXERCISE

1. The expected value of the voltage across a resistor is 15V, however, measurement yields a value of 14.5V.Calculate:
   (a) The absolute error.
   (b) The percent error.
   (c) The relative accuracy.
   (d) The percent accuracy.

2. If the average of set of voltage reading is 30.15V. Calculate the precision of one of the reading that was equal to 29.9V.

3. The following set of measurement was recorded in the laboratory. Calculate the precision of the 4th measurement.

Measurement no.	Measurement value (V)
1	98
2	102
3	101
4	97
5	100
6	103
7	98
8	106
9	107
10	99

4. Calculate the percent of error and percent of accuracy if ammeter at Figure 1.1 shown the reading 1.9A and voltmeter at Figure 1.9 shown the reading 2.9V.

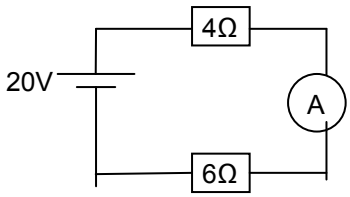


Figure 1.1

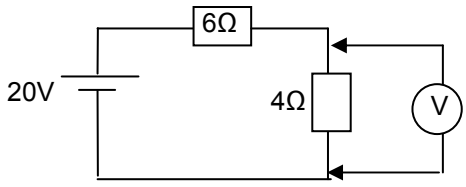


Figure 1.2

5. A ammeter is using to measure current at A. The reading shown 0.75A. Calculate percent of error and percent of accuarcy.

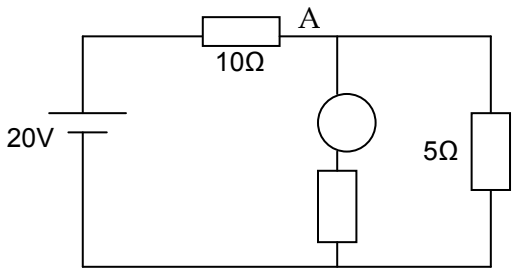


Figure 1.3